Preserving Ariel Reciprocating Compressor Frames, PAG Oil Compatible

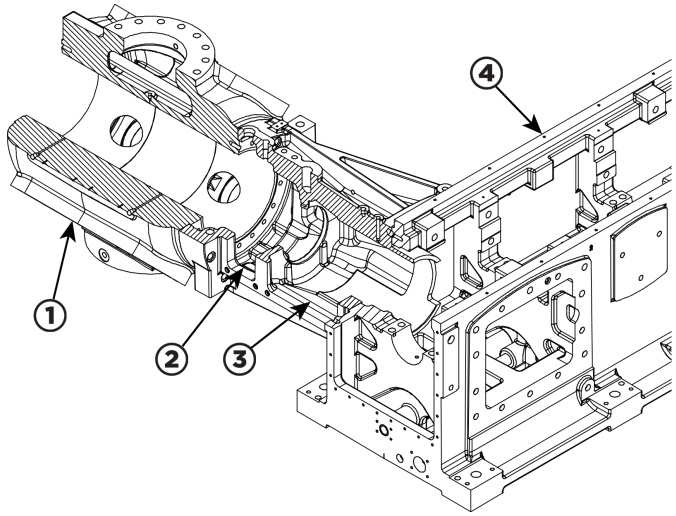
This document pertains to Ariel reciprocating compressor frames preserved at Ariel and/or stored for 12 months since the last preservation. The procedures outlined here help prevent internal corrosion for up to 12 months, provided that the protective measures remain intact. If any signs of deterioration occur or if the crankshaft is turned, re-preserve according to the guidelines in this document.

For new compressors to be stored longer than 12 months from the Ariel shipment date, see ER-10.5.1 and apply for Deferred Start-up Status to defer the standard warranty.

Related Documents:

- ER-34 - Preserving Ariel Non-Lube Cylinders and Distance Pieces with VCI Powder
- ER-129 - Preserving Ariel Reciprocating Compressor Frames, Mineral, Synthetic PAO, and Synthetic Ester Oil Compatible
- ER-130 - Preserving Ariel Cylinders and Distance Pieces, Mineral, Synthetic PAO, and Synthetic Ester Oil Compatible
- ER-141 - Preserving Ariel Cylinders and Distance Pieces, PAG Oil Compatible

Figure: Typical Reciprocating Compressor



1 Cylinder
2 Distance Piece
3 Crosshead Cavity
4 Frame

Preservation Materials and Equipment

> **NOTE:**
>
> Cortec products must be stirred before transferring to a smaller container. After transfer, stir or agitate the container before each use. Keep fluids in sealed containers when not in use.

Use the following or their equivalents:

- Contact rust preventative: Cortec M-528.
- Liquid vapor phase corrosion inhibitor (VCI): Cortec M-528.
- Protective coating: Cortec VpCI-368 (alternative: marine grade grease).
- Waterproof back, UV resistant tape: Shurtape PC667.
- Shipping covers and gaskets to seal cylinder suction and discharge openings, and standard distance piece covers. Install a standard guide cover in place of any cover equipped with an over-pressure relief valve.
- Wax impregnated cloth: Versil-Pak 3-2 JAN-B-121, Type II Grade C, Class II. Available from Daubert Cromwell LLC (Alsip, Illinois).
- Priming pump if the compressor does not have one.
- Tarps and tie down materials for outdoor storage.

Preservation Procedure

1. Inspect and preserve in a clean, dry environment. Remove the following:
   - Top cover.
   - Crosshead guide covers, including any covers equipped with over-pressure relief valves.
   - Tape and wax impregnated cloth from drive stub.
2. Inspect all internal surfaces and cavities.
   a. Inspect for corrosion, standing water, sludge, or gel. If found, photograph affected areas and contact Ariel.
   b. Remove water contamination, corrosion, sludge, or gel prior to coating parts with rust preventative.
   c. Inspect for potential leak paths for water and air and seal them.
   d. Verify metal and plastic plugs are tight and replace damaged plugs.
   e. Inspect exposed stainless steel tubing and verify metal tubing caps are tight.
3. Prime lube oil system with a fresh 50/50 mixture of compressor frame oil and rust preventative.
   a. Pump rust preventative/oil mixture through the lube oil system. Introduce a mixture upstream of the oil filter. Verify that oil visibly flows from all bearing positions and crossheads.
   b. Drain excess rust preventative/oil mixture from frame and guides. Lube oil piping should remain filled with the rust preventative/oil mixture during the storage period.
4. **To prevent damage, perform Step 4 within 8 hours of Step 3.** Thoroughly agitate rust preventative and VCI before each use. Use VCI within 24 months of the manufacture date. Spray all internal compressor surfaces with rust preventative Cortec M-528.
   a. Spray rust preventative inside frame and guide. Cover all internal surfaces.
   b. Rotate crankshaft 180° and repeat Step a. Allow no noticeable accumulation of rust preventative in the bottom of treated cavities.
   c. For models with top and or side covers with o-ring seals: lightly coat the top rail of the frame and side openings of the guide with protective coating before installing top or side covers.
5. Pour, spray, or fog VCI into the frame per Table: Min. Quantity of Cortec M-528 to Preserve Frames. If pouring, distribute evenly throughout the cavity. Do not pour VCI directly on any bearing surfaces. Preserve the crosshead guide compartment in the same way. For distance piece preservation, see ER-141. If your compressor does not appear in Table: Min. Quantity of Cortec M-528 to Preserve Frames, contact Ariel for the correct VCl quantities.
6. Add VCI to the following cavities:
   - Cylinder lube box sump via the fill plug - 1/2 fluid ounce
   - Lube oil filter housings drained prior to shipment (not applicable to spin-on filters) - 1 fluid ounce
   - Lube oil pump discharge port - 1/2 fluid ounce
7. VCI requires protected cavities to remain sealed during storage. Open cavities deplete VCI concentration, rendering it ineffective. To prevent VCI depletion, seal the compressor as follows:
   - Plug all openings, including vents and breather connections. Verify plugs are tight.
   - Reassemble all covers and reconnect the pre-lube pump. Do not install guide covers equipped with relief valves; replace them with standard covers.

Table: Min. Quantity of Cortec M-528 to Preserve Frames - Fluid Oz. (ml)

Frame	Min. Quantity
JGM:P:N:Q:I, KB100, JG:JGA/2, JGR:J/2	3 (89)
JG:JGA/4, JGH:E:K:T/2, KBE:K:T/2	6 (177)
JGA/6, JGR:J/4, JGC:D:F/2, KBC:D:F/2	10 (296)
JGJ/6, JGH:E:K:T:C:D:F/4,JGZ:U/2, KBE:K:T:C:D:F/4, KBZ:U/2	16 (473)
JGH:E:K:T:C:D:F/6, JGZ:U/4, KBK:T:C:D:F/6, KBZ:U/4	22 (651)
JGB:V/4, JGZ:U/6, KBZ:U/6, KBB:V/4	36 (1065)
JGB:V/6, KBB:V/6	50 (1479)

8. Coat drive stub with rust preventative, wrap with wax impregnated cloth, followed by waterproof backed tape.
9. Tape the clearance opening between the drive end dust seal cover and crankshaft to prevent any exchange of air to the frame.
10. Ensure force feed pump stroke adjustment lock nuts are wrench tight, and the priming stem is covered with wax impregnated cloth.
11. Ensure all vents, drains, and process piping are capped, plugged, or sealed to prevent air from entering the compressor.
12. Tag the compressor and guide/cylinder assemblies with a tag stating that the crankshaft should not be turned and the date of the last inspection and preservation. If the crankshaft is turned for any reason, re-preserve according to the guidelines in this document.

Storage

Prior to packaging, store in a clean, dry environment. If stored outdoors, protect from direct contact with the elements. Never store with the mounting feet in direct contact with the ground; protect them from corrosion. If stored after packaging, protect all piping and bottles in the package as well. Verify all openings in the package are sealed.

Commissioning Compressor to Service

1. Inspect all compressor internal surfaces and cavities. Inspect for corrosion, standing water, sludge, or gel. If found, photograph affected areas and contact Ariel.
2. Change the oil filter. There is no need to change oil, or if drained, to flush the frame prior to filling with oil; rust preventative and VCI will mix with the lube oil. For PAG synthetic oil applications, consult Ariel.
3. Remove temporary covers, gaskets, plugs, tape, tags, etc. Replace all connections, including vents and drains. The installation guide covers equipment equipped with relief valves, if applicable.
4. Complete the Start-Up Check list (ER-10.6).

CHANGE DESCRIPTION: 1st ¶: 'preserved at Ariel and/or' WAS 'that have been'.